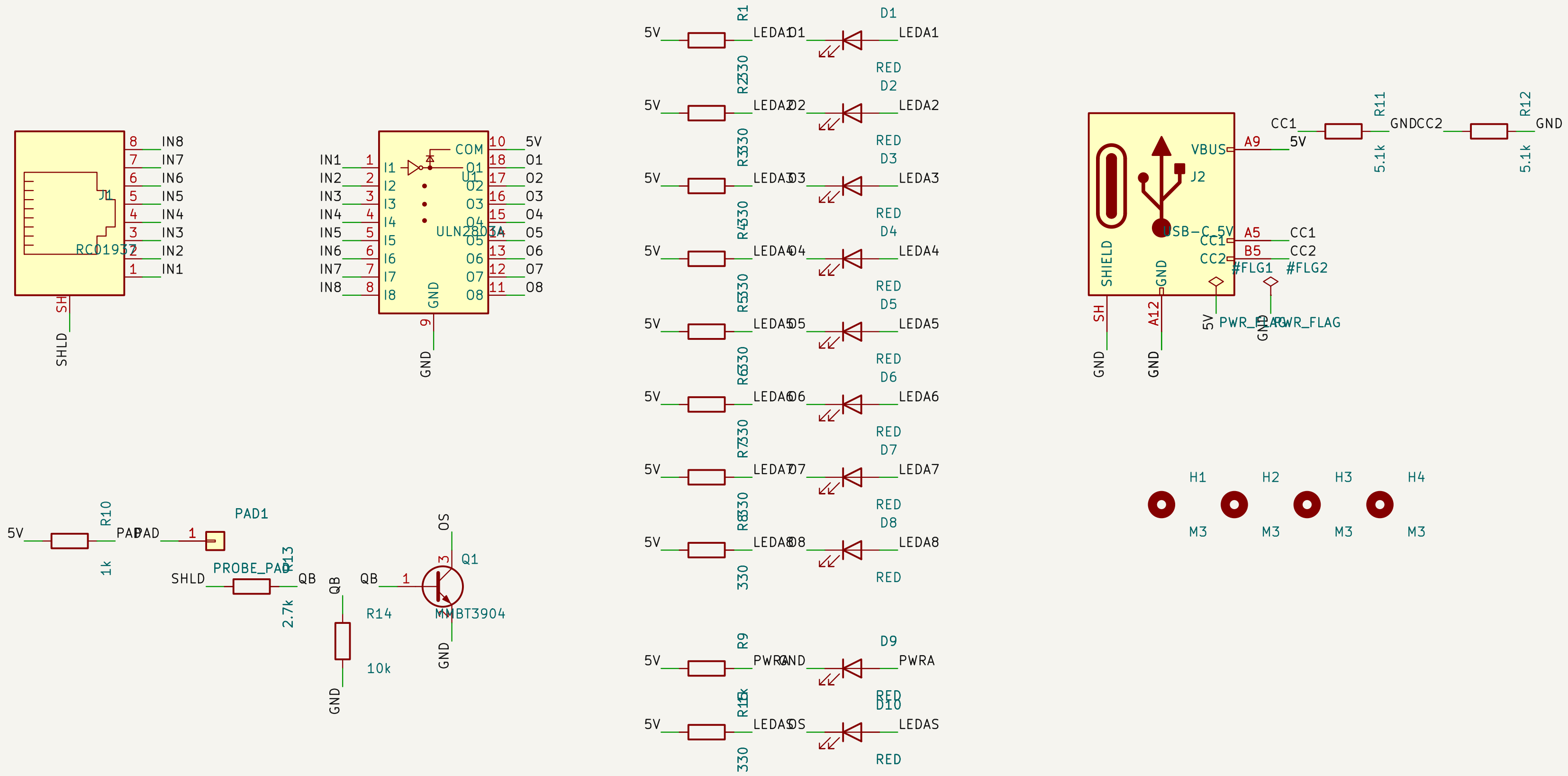
RJ45 CABLE MAPPER rev A rixnobis

Touch a flying wire to PAD1; the LED for its 8P8C pin lights. ULN2803A sinks each LED. PAD at 5V through 1k, so a wire grounded at the far end draws 5 mA, not a short. A finger bridging pad and wire (~1 Mohm) leaves the ULN input far below its ~2.7 V on threshold, so only metal–to–metal contact lights an LED. Two shorted wires light two.



RJ45 jack (RCH RC01937, LCSC C708652) is the only through–hole part, hand–soldered. Everything else is SMD, JLC–assembled. The jack is shielded, so its shell is a 9th channel (Q1, LED S): a shielded cable's drain wire lights S.

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